PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

PHYSICS FORM I REGIONAL EXAMINATIONS

: PHYSICS MARKING SCHEMES

QN1

i	ii	iii	iv	v	vi	vii	viii	ix	x	xi	xii	xiii	xix	xv	xvi
C	B	D	C	B	B	B	C	B	B	B	A	B	D	C	B

xvii	xviii	xix	xx											
B	B	A	D											

(1mark each) Total = 20marks

QN2

LISTA	i	ii	iii	iv	v	vi	vii	viii
LISTB	K	D	A	E	P	G	J	M

(1mark each) Total = 8marks

3(i) Laboratory

(ii) Fundamental forces and Non-fundamental forces

(iii) 0.001cm and 0.01cm

(iv) Data given

Mass, m = 2.5kg

weight = ?

g = 10N/kg

W = mg

= 2.5kg x 10N/kg

W = 25N

∴ weight = 25N

(v) Parallax error

(01mark each) total mark = 5

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Mass	Weight

(i)

Mass	Weight
- Is the quantity of matter in an object - Its SI unit is kilogram - It is measured by beam balance - It is fundamental quantity	- Is the force of attraction that earth exert on a body - Its SI unit is Newton - It is measured by spring balance - It is derived quantity

(0.5mark each) total = 4mark

(ii)

Force	Volume
- Is the pull or push experienced by an object - Its SI unit is Newton	- Is the amount of space occupied by an object - Its unit is cubic metre

(0.5mark each) total = 4marks

(iii)

Sinking	Floating
Is the tendency of an object to fall to lower levels in fluid	Is the tendency of an object to be suspended in (remain) on the surface of fluid due to upthrust

(0.5 mark each) total = 02mark

(iv)

Hydrometer	Measurement
- Is the instrument used for measuring density and relative density of liquid	- Is the process of assigning number to observations or events

(0.5 each) total = 02mark

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5(a)(i) Friction is the property of a surface to oppose sliding of another body on it

1mark

(ii)
- It deforms bodies
- It changes direction of a body moving
- It results to a change of position of a body
- It causes work to be done

(0.5 each) total = 2marks

(b) Density is the mass per unit volume of a substance

$\rho = m/v$ (1mark)

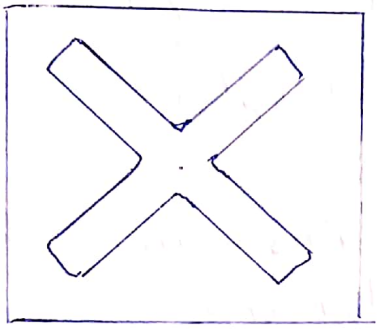
ii) Applications of density
- In designing and building ships
- Operating submarines
- In operation of hydrometers
- In designing aircrafts

(2mark total)

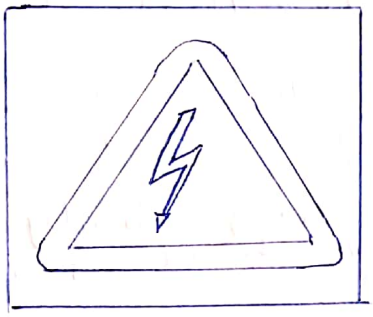
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a) First Aid is an immediate assistance or care given to a sick or injured person before getting Professional medical help (1mark)

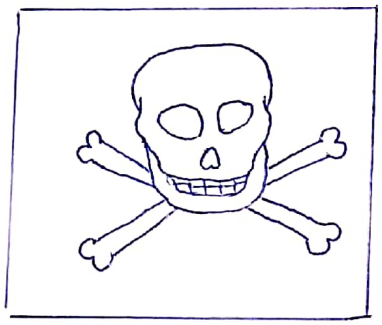
b

(iv) 

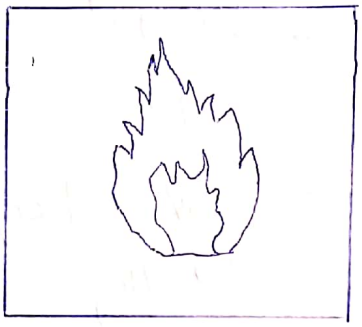
Irritant

(ii) 

Danger of electric shock

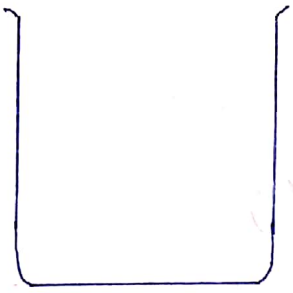
(i) (iii) 

TOXIC

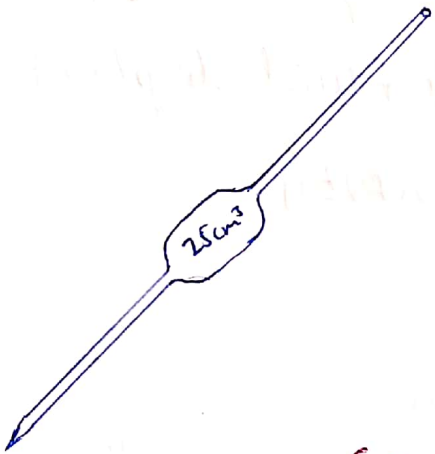
(iv) 

Flammable

(c)

(i) 

Beaker

(ii) 

Pipette

(2.0 mark each) total = 12 marks

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7(a) Archimedes' principle state that "When a body is totally or partially immersed in a fluid it experience an upthrust which is equal to the weight of fluid displaced by the body" 1½ mark

- "A floating body displaces its own weight in which it floats" 1½ mark

(b) Soln

(i) Mass of body

$$= \frac{\text{Weight}}{10} \quad \text{0/mark}$$

$$= \frac{10N}{10N/kg} = 1kg$$

Mass of fluid displaced

$$= \frac{\text{upthrust}}{10}$$

$$= \frac{10-8}{10} = 0.2kg \quad \text{0/mark}$$

∴ Mass of fluid displaced = 0.2kg

Volume of fluid displaced = $\frac{\text{Mass of fluid displaced}}{\text{Density}}$ 1mark

$$= \frac{0.2kg}{0.89/cm^3}$$

$$= \frac{200g}{0.89/cm^3} \quad \text{0.5 marks}$$

∴ Volume of fluid displaced = $250 cm^3$ 1mark

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Condition for floating

- Upthrust exerted by fluid is less than weight of object
- object

→ Upthrust exerted by fluid must be equal to the real weight of the object or greater to the real weight of the object

→ Density of body must be less than that of fluid

→ Volume of submerged object must be large enough to displace a lot of fluid (1mark each) total = 3marks

QN9

(a) Error is the difference between true value and measured value (1mark)

(b) (i) 400mm into m

Soln

From

$1m = 1000mm$ 0.5mark

$y = 400MM$

$y = \frac{1m \times 400M}{1000mm}$ 0.5mark

$= 0.4m$

$\therefore 400mm = 0.4m$ (1mark)

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7(b)

(ii) Density of the body

$$= \frac{\text{Mass of the body}}{\text{Volume displaced}}$$ 0½ mark

$$= \frac{1000g}{250cm^3}$$ 0.5mark

$$= 4g/cm^3$$

∴ Density of the body

$= 4g/cm^3$ 0½mark

8(a) - Hospital
- Industry
- School

(0.5mark each) total = 1.5marks

(b) Data given

Mass, M = 500N

gravitation, g = 10N/kg (0½mark)

Acceleration, a = ?

From formula:

$F = ma$

$a = F/M$ (0.5mark)

$= \frac{500N}{10N/kg}$ (0.5mark)

$= 50m/s^2$

$a = 50m/s^2$

or 50N/kg 1mark

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(ii) 2.5km — m

Recall

1km = 1000m (0.5 mark)

2.5km = a

$$a = \frac{2.5\text{km} \times 1000\text{m}}{1\text{km}}$$ (0.5 mark)

$$= 2500\text{m}$$

∴ 2.5km = 2500m (0 mark)

(iii) 65min → sec

Recall

1min = 60sec (0.5 mark)

65min = y

$$y = \frac{65\text{min} \times 60\text{sec}}{1\text{min}}$$ (0.5 mark)

$$= 3900\text{seconds}$$

∴ 65min = 3900seconds (0 mark)

(iv) 2hr → sec

From

1hr = 3600sec (0.5 mark)

2hr = x

$$x = \frac{2\text{hr} \times 3600\text{sec}}{1\text{hr}}$$ (0.5 mark)

$$= 7200\text{seconds}$$

∴ 2hr = 7200seconds 0 mark

(v) 126g → kg

From

1kg = 1000g (0.5 mark)

y = 126g

$$y = \frac{126\text{g} \times 1\text{kg}}{1000\text{g}} = 0.126\text{kg}$$ (0.5 mark)

∴ 126g = 0.126kg (0 mark)

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10(a) Relative density is the ratio of density of substance to density of water

$$R.D = \frac{\text{Density of substance}}{\text{Density of water}}$$ (1mark)

(b)

Data given

Mass of metal, $M_m = 0.72g$ 0.5mark

Volume of metal, $V_m = 0.24cm^3$

R.d is required = ?

From

$\rho = M/V$ (0.5 mark)

$= 0.72g/0.24cm^3$ 0.5mark

$\rho_m = 3g/cm^3$ 1mark

Now, R.d of metal

$= \rho_m/\rho_w$ (0.5 mark)

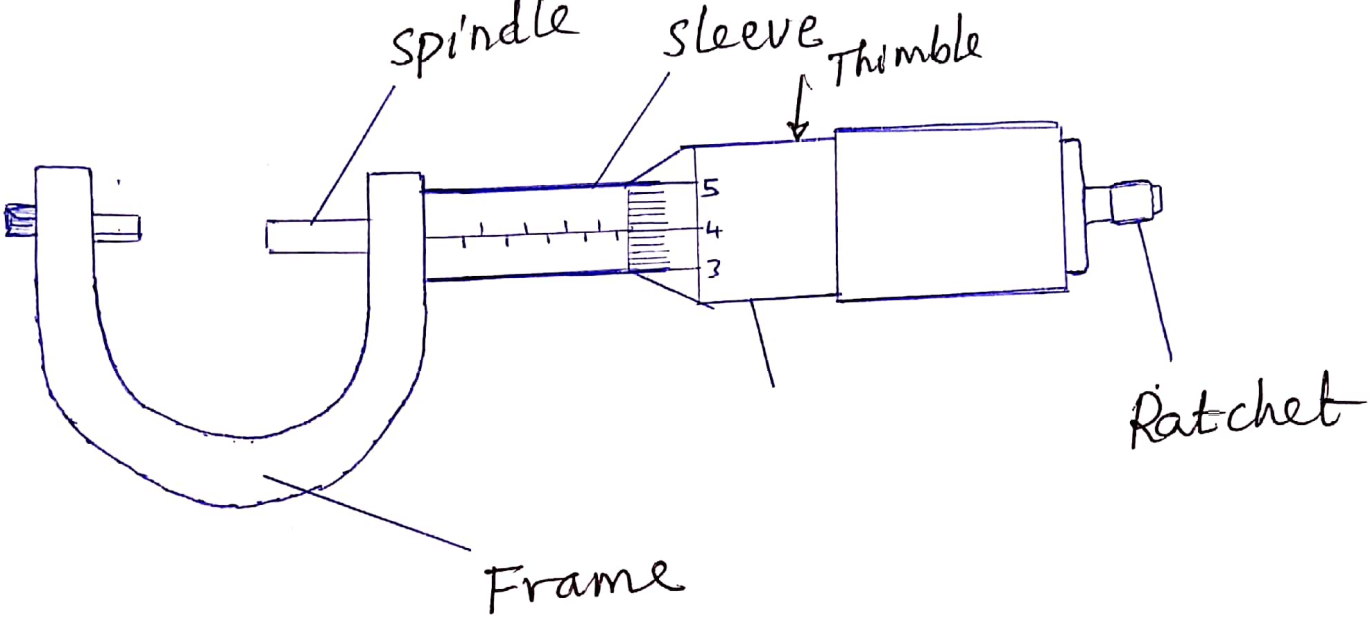
$= \frac{3g/cm^3}{1g/cm^3}$ (0.5mark)

$= 3$

∴ Relative density of metal

$= 3$ (0.5mark)

Diagram of Micrometer screw gauge



(4 marks) total mark

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